

CLOUD SECURITY · AWS IAM

Assignment 18

Multi-Factor Authentication (MFA)

A Clean, Structured Study Guide

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Topic: Enabling, Testing & Comparing MFA on AWS IAM

Platform: Amazon Web Services (AWS)

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What is MFA?

DEFINITION

Multi-Factor Authentication adds a second layer of security using a 6-digit code in addition to the password.

Why is MFA important?

- Protects against stolen passwords
- Prevents unauthorized logins
- Recommended by AWS Security Best Practices

TIP

Even if a password leaks through phishing, MFA stops the login cold — the attacker still needs the rotating 6-digit code from the user's authenticator app.

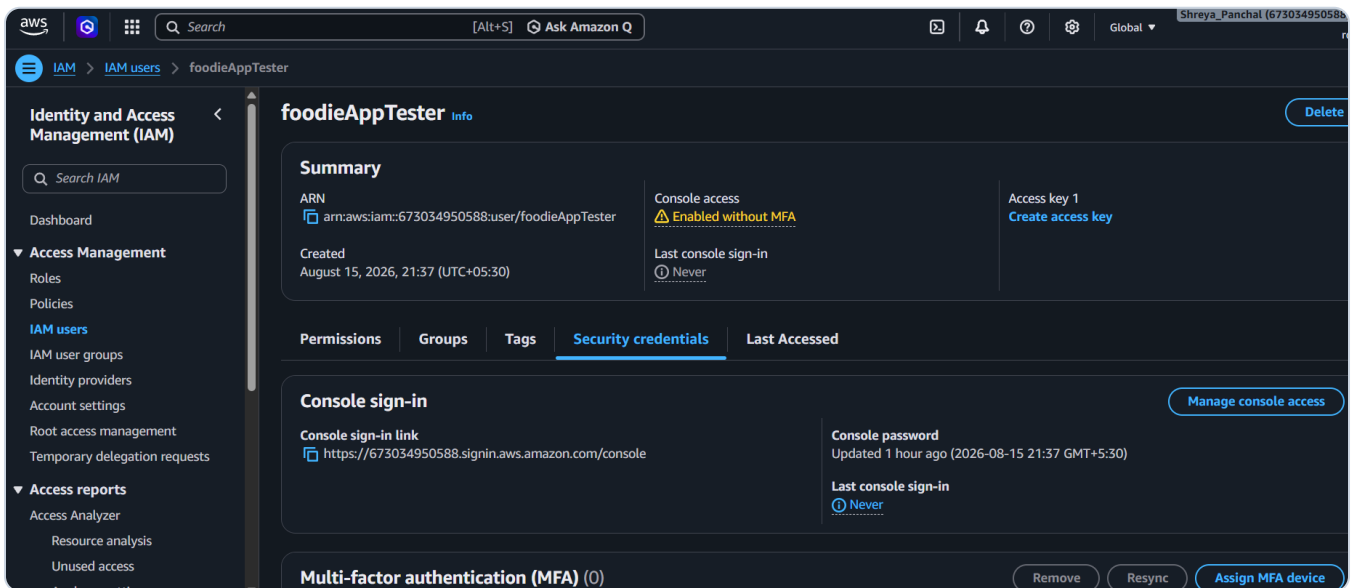
Task 1 • Enable MFA on AWS IAM User

TARGET USER

foodieAppTester

Step 1: Open Security Credentials

AWS Console → IAM → Users → foodieAppTester → Security credentials



Screenshot: IAM → Users → foodieAppTester → Security credentials

Step 2: Assign MFA

Scroll down to Multi-factor authentication (MFA) → Click Assign MFA device

Step 3: Select Device

Choose: Authenticator app (Recommended)

Click Next

Step 4: Install Authenticator

Phone me koi bhi app install karo:

- Google Authenticator
- Microsoft Authenticator
- Authy

Step 5: Scan QR Code

AWS ek QR code dikhayega.

1. Authenticator app open karo
2. + → Scan QR Code
3. 6-digit code generate hoga

Step 6: Verify

AWS me:

- MFA Code 1 → Enter current 6 digits
- Wait 30 sec
- MFA Code 2 → Enter next 6 digits

Click Add MFA

EXPECTED RESULT

Status: **Assigned**

SCREENSHOT TO CAPTURE

MFA assigned confirmation page.

Task 2 • Test Login with MFA

Incognito Browser

1. Open Chrome Incognito
2. Visit AWS Sign-in URL
3. Enter: Account ID, Username: `foodieAppTester` , Password
4. New screen ayegi → Enter 6-digit MFA code
5. Login successful

Document These Steps

Step	Without MFA	With MFA
Username	Yes	Yes
Password	Yes	Yes
6-digit Code	No	Yes
Login	Direct	Verified

SCREENSHOT TO CAPTURE

MFA code prompt.

Task 3 • Disable MFA & Compare

Disable MFA

IAM → Users → foodieAppTester → Security credentials

Under MFA:

- Deactivate
- Confirm

Now login again.

Comparison Table

Feature	Without MFA	With MFA
Password Only	Yes	No
6-digit Code	No	Yes
Security	Low	High
Hacker Protection	Weak	Strong

SCREENSHOT TO CAPTURE

Take screenshot of Security Credentials after deactivation if your instructor requires it.

NOTE

Agar ye personal account hai, assignment complete hote hi MFA dubara Enable kar dena.

Task 4 • Scenario (3–5 Lines)

Answer

An attacker steals a user's AWS username and password through phishing. Without MFA, the attacker can log in and access cloud resources. If MFA is enabled, the attacker must also enter the **6-digit verification code** from the user's authenticator app. This blocks unauthorized access and helps protect the AWS account.